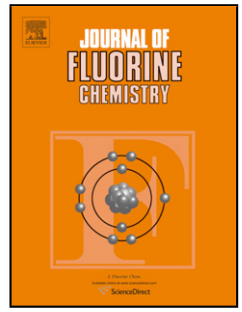
Accepted Manuscript



Title: A Versatile Difluorovinylation Method: Cross-Coupling Reactions of the 2,2-Difluorovinylzinc–TMEDA Complex with Alkenyl, Alkynyl, Allyl, and Benzyl Halides

Author: Tomohiro Ichitsuka Tsuyoshi Takanohashi Takeshi Fujita Junji Ichikawa

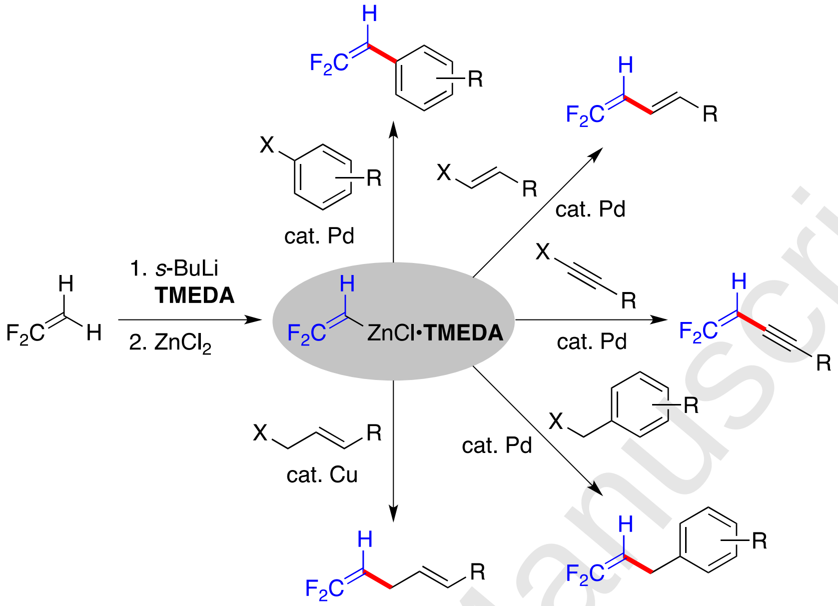
PII: S0022-1139(14)00366-2
DOI: http://dx.doi.org/doi:10.1016/j.jfluchem.2014.12.003
Reference: FLUOR 8480

To appear in: *FLUOR*

Received date: 1-12-2014
Revised date: 6-12-2014
Accepted date: 8-12-2014

Please cite this article as: T. Ichitsuka, T. Takanohashi, T. Fujita, J. Ichikawa, A Versatile Difluorovinylation Method: Cross-Coupling Reactions of the 2,2-DifluorovinylzincndashTMEDA Complex with Alkenyl, Alkynyl, Allyl, and Benzyl Halides, *Journal of Fluorine Chemistry* (2014), http://dx.doi.org/10.1016/j.jfluchem.2014.12.003

Graphical Abstract (Pictogram)



Graphical Abstract (Synopsis)

A series of 2,2-difluorovinyl compounds were synthesized via the Pd- and Cu-catalyzed cross-coupling of a 2,2-difluorovinylzinc–TMEDA complex. The 2,2-difluorovinylzinc–TMEDA complex, prepared from 1,1-difluoroethylene, an industrial material, is thermally stable and storable.

Highlights

A series of 2,2-difluorovinyl compounds were synthesized via the cross-coupling of a 2,2-difluorovinylzinc–TMEDA complex.>This protocol started from an industrial material, 1,1-difluoroethylene..>The 2,2-difluorovinylzinc–TMEDA complex is thermally stable and storable.

A Versatile Difluorovinylation Method: Cross-Coupling Reactions of the 2,2-Difluorovinylzinc–TMEDA Complex with Alkenyl, Alkynyl, Allyl, and Benzyl Halides

Tomohiro Ichitsuka, Tsuyoshi Takanohashi, Takeshi Fujita, Junji Ichikawa*

Division of Chemistry, Faculty of Pure and Applied Sciences, University of Tsukuba,

Tsukuba, Ibaraki 305-8571, Japan

*Corresponding author. Fax: +81-29-853-4237
E-mail address: junji@chem.tsukuba.ac.jp (J. Ichikawa).

Dedicated to Professor Iwao Ojima on the occasion of his 70th birthday.

Abstract

A thermally stable 2,2-difluorovinylzinc–TMEDA complex was prepared via a deprotonation–transmetallation sequence starting from commercially available 1,1-difluoroethylene. The complex thus formed was successfully applied to transition metal-catalyzed cross-coupling reactions with a wide range of organic halides, which led to the syntheses of 2,2-difluorovinyl compounds. On treatment with the difluorovinylzinc–TMEDA complex in the presence of an appropriate palladium or copper catalyst, alkenyl, alkynyl, allyl, and benzyl halides effectively underwent difluorovinylation to afford 1,1-

difluoro-1,3-dienes, 1,1-difluoro-1,3-enynes, 1,1-difluoro-1,4-dienes, and (3,3-difluoroallyl)arenes, respectively.

Keywords: Difluoroalkene, Organozinc, Cross-Coupling Reaction, Palladium, Copper, Catalyst

1. Introduction

2,2-Difluorovinyl compounds are an important class of compounds because they exhibit unique properties due to the steric and electronic effects of fluorine. They serve as not only building blocks for fluorine-containing organic molecules [1] but also monomers for functional polymers [2]. In addition, 2,2-difluorovinyl compounds often show substantial bioactivities. For example, they act as anti-herpes simplex virus type 1 (anti-HSV-1) agents [3] and as squalene epoxidase inbibitors in antilipemic drugs [4]. Further pharmaceutical applications of difluorovinyl compounds have been of great interest, since the difluorovinylidene moiety is considered to be a bioisostere of a carbonyl group [5].

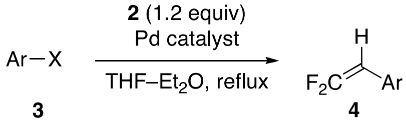
Despite the usefulness of 2,2-difluorovinyl compounds, their availability is still limited. Typical synthetic methodologies are mostly classified into two categories: (i) difluoromethylenation of aldehydes and (ii) metal-mediated difluorovinyl cross-coupling. The former protocol involves the Wittig reaction [6], the Horner–Wadsworth–Emmons reaction [7], and the Julia–Kocienski reaction [8] with aldehyde substrates. Although these reactions are widely used in common alkene synthesis, the Wittig reaction requires excess amounts of intermediary ylides, and the Horner–Wadsworth–Emmons and Julia–Kocienski reactions show narrow substrate scopes due to the necessity of highly basic conditions. Alternatively, the latter metal-mediated cross-coupling has been considered to be a more straightforward approach to difluorovinyl compounds (Scheme 1a). This protocol is achieved via the reaction between 2,2-difluorovinyl halides and organometallic species [9] or the reaction between 2,2-difluorovinylmetals and organic halides [10]. Both types of reactions require starting difluorovinyl halides, which are expensive or rarely available from commercial sources.

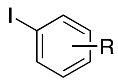
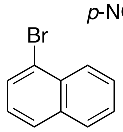
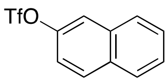
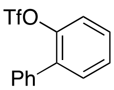
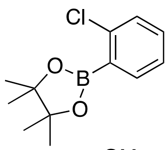
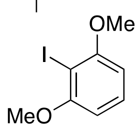
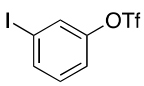
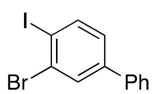
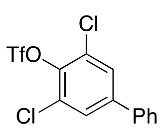
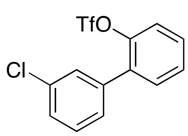
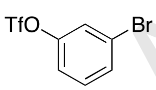
Normant *et al*. reported the synthesis of a 2,2-difluorovinyl compound starting from 1,1-difluoroethylene (**1**), a commercially available, industrial material (Scheme 1b) [11]. In the study, a difluorovinylzinc complex, prepared via the deprotonation of **1** and subsequent transmetallation, was subjected to a palladium-catalyzed cross-coupling reaction with 2-iodopyridine. This is the only reported cross-coupling reaction with the difluorovinylzinc complex derived from 1,1-difluoroethylene, and the product yield was no more than 50%, which was presumably due to the thermal instability of the intermediary zinc complex. Thus, difluorovinylation via a cross-coupling reaction remains to be developed in terms of both generality and efficiency.

To enhance the versatility of this type of reaction, we sought the appropriate ligand that stabilized the difluorovinylzinc complex [12,13]. Finally, we found that the coordination of *N*,*N*,*N′*,*N′*-tetramethylethylenediamine (TMEDA) effectively stabilized the complex (Scheme 1c). The resulting 2,2-difluorovinylzinc–TMEDA complex **2** was storable and was successfully applied to the palladium-catalyzed cross-coupling reaction with aryl halides **3** [14], affording β,β-difluorostyrenes **4** in high yields with high chemoselectivity (I > OTf > Br > Cl, Table 1) [12]. Herein, we report the palladium- or copper-catalyzed cross-coupling reactions of the difluorovinylzinc–TMEDA complex **2** with alkenyl, alkynyl, benzyl, and allyl halides; the reactions provide facile syntheses of 1,1-difluoro-1,3-dienes **6**, 1,1-difluoro-1,3-enynes **8**, (3,3-difluoroallyl)arenes **10**, and 1,1-difluoro-1,4-dienes **12**, respectively (Scheme 1c).

Scheme 1. Syntheses of difluorovinyl compounds via transition-metal-catalyzed cross-coupling

Table 1. Synthesis of β,β-difluorostyrenes **4** by Pd-catalyzed cross-coupling of **2** with aryl halides **3** [12]



entry	3			X	Pd catalyst (mol%)	Time (h)	Yield (%)[a]
1		*p*-Me	**3a**	I	$Pd(PPh_3)_4$ (2)	6	**4a** 59 (86)
2		*p*-OMe	**3b**	I	$Pd(PPh_3)_4$ (2)	6	**4b** 87
3		*o*-NH_2	**3c**	I	$Pd(PPh_3)_4$ (2)	1	**4c** 87
4		*p*-NO_2	**3d**	I	$Pd_2(dba)_3$ (2)/ PPh_3 (8)	12	**4d** 84
5			**3e**	Br	$Pd(PPh_3)_4$ (2)	10	**4e** 87
6			**3f**	OTf	$Pd(PPh_3)_4$ (2)	6	**4f** 90
7			**3g**	OTf	PEPPSI-IPr (5)	10	**4g** 82
8			**3h**	Cl	PEPPSI-IPr (5)	6	**4h** 73
9			**3i**	I	PEPPSI-IPr (4)	12	**4i** 79
10			**3j**	I	$Pd_2(dba)_3$ (2.5)/ P(2-furyl)$_3$ (10)	4	**4j** 92
11			**3k**	I	$Pd_2(dba)_3$ (2.5)/ P(2-furyl)$_3$ (10)	4	**4k** 96
12			**3l**	OTf	$PdCl_2$(dppp) (5)	23	**4l** 73
13			**3m**	OTf	$PdCl_2$(dppf) (4)	12	**4m** 87
14			**3n**	OTf	$Pd_2(dba)_3$ (2.5)/ dppb (5)	24	**4n** 69

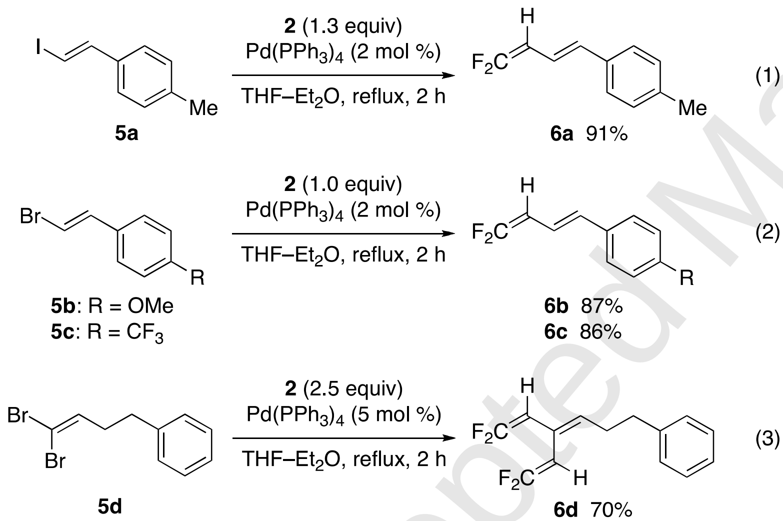
[a] Isolated yield. ^{19}F NMR yield using $PhCF_3$ as an internal standard is indicated in parentheses.

2. Results and discussion

2.1. Cross-coupling reaction with alkenyl halides: Synthesis of 1,1-difluoro-1,3-dienes

The optimized conditions for the reactions of the difluorovinylzinc complex **2** with aryl halides **3** were successfully applied to the reactions with alkenyl halides **5** (eqs. 1–3). On treatment with 1.3 equiv.

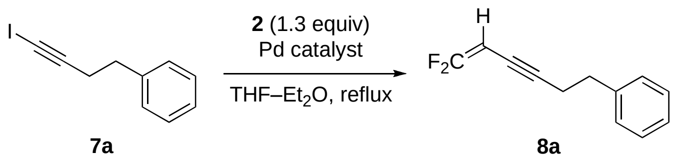
of **2** in the presence of 2 mol % of $Pd(PPh_3)_4$, (*E*)-β-iodo-*p*-methylstyrene (**5a**) smoothly underwent a cross-coupling reaction to afford 1,1-difluoro-1,3-dienes **6a** in 91% yield (eq. 1). In this reaction, the *E* configuration of the alkenyl moiety was definitely retained. β-Bromostyrenes **5b** and **5c**, bearing electron-donating and electron-withdrawing substituents, also participated in the cross-coupling reaction to afford the corresponding 1,1-difluoro-1,3-dienes **6b** and **6c**, respectively, in high yields with the retention of the *E* configuration (eq. 2). Note that the double difluorovinylation of 1,1-dibromo-1-alkene **5d** effectively proceeded to afford the tetrafluorinated, cross-conjugated triene ([3]dendralene) **6d** in 70% yield (eq. 3).



2.2. Cross-coupling reaction with alkynyl halides: Synthesis of 1,1-difluoro-1,3-enynes

When cross-coupling the difluorovinylzinc complex **2** with alkynyl halides **7**, the choice of ligand for palladium was critical (Table 2). The use of $Pd(PPh_3)_4$ as a catalyst in the reaction of **2** with alkynyl iodide **7a** gave 1,1-difluoro-1,3-enyne **8a** as the expected product, albeit in a moderate yield of 59% (entry 1). Some bidentate phosphine ligands were found to improve the yield of **8a** (entries 3–7). Among the bidentate ligands examined, 1,3-bis(diphenylphosphino)propane (dppp) afforded the highest yield of **8a**, 96% (entry 5), while 1,1'-bis(diphenylphosphino)ferrocene (dppf) also gave a satisfactory yield of 86% (entry 7).

Table 2. Effect of ligands for Pd-catalyzed cross-coupling of **2** with alkynyl halide **7a**

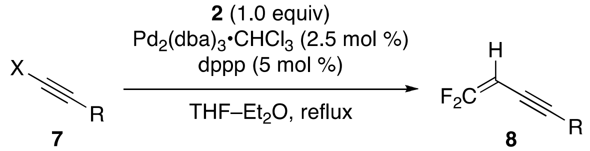


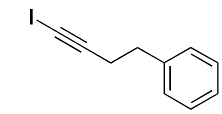
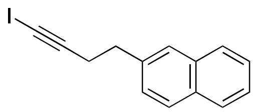
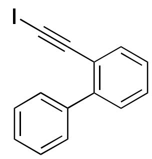
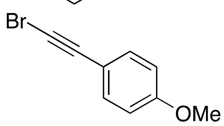
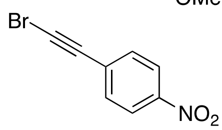
entry	Pd catalyst (mol %)	Time (h)	Yield (%)[a]
1	$Pd(PPh_3)_4$ (4)	3	59
2	$Pd_2(dba)_3{\bullet}CHCl_3$ (2)/ PPh_3 (8)	4	66
3	$Pd_2(dba)_3{\bullet}CHCl_3$ (2.5)/ dppm (5)	5	27
4	$Pd_2(dba)_3{\bullet}CHCl_3$ (2.5)/ dppe (5)	2	67
5	$Pd_2(dba)_3{\bullet}CHCl_3$ (2.5)/ dppp (5)	5	96
6	$Pd_2(dba)_3{\bullet}CHCl_3$ (2.5)/ dppb (5)	5	68
7	$Pd_2(dba)_3{\bullet}CHCl_3$ (2.5)/ dppf (5)	5	86

[a] ^{19}F NMR yield using $PhCF_3$ as an internal standard.

With optimized conditions in hand, the substrate scope was investigated (Table 3). Along with alkyl-substituted ethynyl iodides **7a** and **7b** (entries 1 and 2), aryl-substituted ethynyl iodide **7c** effectively underwent the Pd(0)/dppp-catalyzed cross-coupling reaction with **2** to afford the corresponding 1,1-difluoro-1,3-enyne **8c** (entry 3). Difluorovinylation of alkynyl bromides was also successfully achieved under similar conditions (entries 4 and 5). The reaction of arylethynyl bromide **7d**, bearing an electron-donating methoxy group, effectively afforded 1,1-difluoro-1,3-enyne **8d** in 94% yield (entry 4), while the cross-coupling of alkynyl bromide **7e**, with an electron-withdrawing nitro group, afforded enyne **8e** in 63% yield (entry 5).

Table 3. Synthesis of 1,1-difluoro-1,3-enynes **8** by Pd-catalyzed cross-coupling of **2** with alkynyl halides **7**



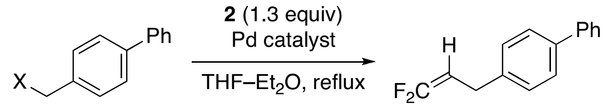
entry	7		Time (h)	Yield (%)[a]
1[b]		**7a**	5	**8a** 85
2		**7b**	2	**8b** 94
3[c]		**7c**	5	**8c** 90
4		**7d**	5	**8d** 94
5		**7e**	1	**8e** 63

[a] Isolated yield. [b] **2** (1.3 equiv). [c] **2** (1.2 equiv).

2.3. Cross-coupling reaction with benzyl halides: Synthesis of (3,3-difluoroallyl)arenes

Cross-coupling reactions of the difluorovinylzinc complex **2** with benzyl bromides were troublesome because unavoidable self-coupling led to dibenzyls (Table 4). In the presence of 5 mol % of $Pd(PPh_3)_4$, the reaction of **2** with 4-phenylbenzyl bromide (**9a**) afforded a 27% yield of 4-(3,3-difluoroallyl)biphenyl **10a**, where the rest of **9a** was mostly converted to its homocoupling product (entry 1). Addition of sodium iodide provided a slightly better result, a 39% yield of **10a** (entry 2). No effective catalyst was found on screening the ligands (entries 3–7). These results indicated that benzyl bromide **9a** was too reactive to be used as a substrate for palladium-catalyzed cross-coupling with **2**. Eventually, we found that the use of 1.0 equiv. of 4-phenylbenzyl chloride (**9'a**) drastically improved the yield of **10a**, up to 92%, by suppressing the formation of the homocoupling product (entry 8).

Table 4. Effect of conditions for Pd-catalyzed cross-coupling of **2** with benzyl halides **9a** and **9'a**



X = Br: **9a**
X = Cl: **9'a**

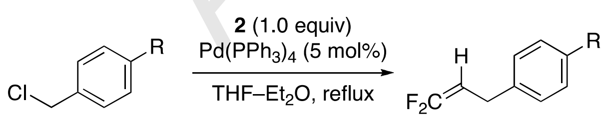
10a

entry	X	Pd catalyst (mol%)	Time (h)	Yield (%)[a]
1	Br	$Pd(PPh_3)_4$ (5)	1	27
2	Br	$Pd_2(dba)_3\cdot CHCl_3$ (2.5)/ NaI (130)	1	39
3	Br	$Pd_2(dba)_3\cdot CHCl_3$ (2.5)/ PPh_3 (10)	2	19
4	Br	$Pd_2(dba)_3\cdot CHCl_3$ (2.5)/ PCy_3 (10)	1	3
5	Br	$Pd_2(dba)_3\cdot CHCl_3$ (2.5)/ P(*t*-Bu)$_3$ (5)	2	33
6	Br	$Pd_2(dba)_3\cdot CHCl_3$ (2.5)/ dppe (5)	2	N.D.
7	Br	PEPPSI-IPr (5)	1	14
8[b]	Cl	$Pd(PPh_3)_4$ (5)	2	92

[a] ^{19}F NMR yield using $PhCF_3$ as an internal standard. [b] **2** (1.0 equiv).

The synthesis of several (3,3-difluoroallyl)arenes **10** was examined via the cross-coupling of **2** with benzyl chlorides **9'** (Table 5). Benzyl chlorides **9'b**–**9'd** bearing electron-donating (entries 2 and 3) and electron-withdrawing substituents (entry 4) successfully underwent palladium-catalyzed difluorovinylation using **2** to afford the corresponding (3,3-difluoroallyl)arenes **10b**–**10d**, respectively.

Table 5. Synthesis of (3,3-difluoroallyl)benzenes **10** by Pd-catalyzed cross-coupling of **2** with benzyl chlorides **9'**



9'

10

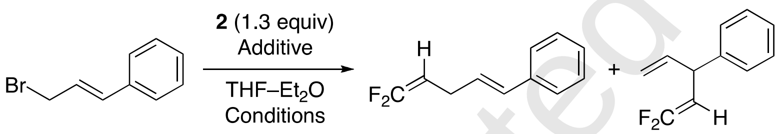
entry	R		Time (h)	Yield (%)[a]
1	R = Ph	**9'a**	2	**10a** 92
2[b]	R = Bu	**9'b**	2	**10b** 93
3	R = OMe	**9'c**	2	**10c** 82
4	R = CF_3	**9'd**	6	**10d** 65 (73)

[a] Isolated yield. ^{19}F NMR yield using $PhCF_3$ as an internal standard is indicated in parentheses. [b] **2** (1.3 equiv).

2.4. Cross-coupling reaction with allyl halides: Synthesis of 1,1-difluoro-1,4-dienes

We finally investigated the cross-coupling of the difluorovinylzinc complex **2** with allylic halides, which have two possible reaction sites, namely, carbons α and γ to the leaving halo group (Table 6). In the difluorovinylation using allylic halide *E*-**11a** as a model substrate, the palladium complexes such as $Pd(PPh_3)_4$ exhibited poor catalyst efficiencies (entry 1). Addition of more than a stoichiometric amount of CuI instead of the Pd(0) catalyst significantly improved the yield of difluorovinylated products, with the S_N2-type product *E*-**12a** preferentially obtained along with S_N2'-type product **13a** (entries 2 and 3). Among the Cu(I) species examined, a catalytic amount of $CuBr{\cdot}SMe_2$ afforded the highest yield of difluorovinylated products and the highest selectivity in the formation of *E*-**12a** (*E*-**12a**/**13a** = 92:8, entry 6).

Table 6. Effect of conditions for Cu-catalyzed cross-coupling of **2** with allylic halide *E*-**11a**

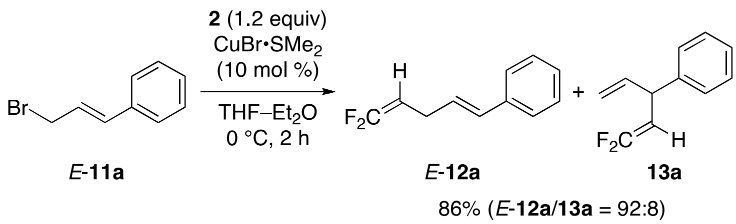


11a *E*-**12a** **13a**

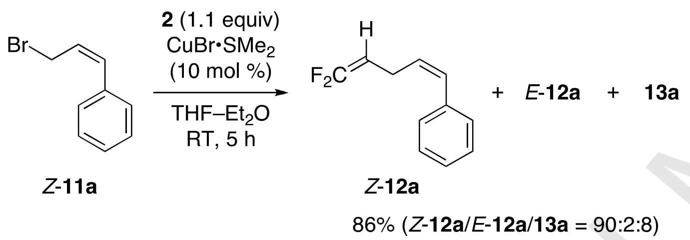
entry	Additive	Conditions	Yield (%)[a]	*E*-**12a**/**13a**[b]
1	$Pd(PPh_3)_4$ (2 mol%)	reflux, 3 h	5	–
2	CuI (1.3 equiv)	RT, 2 h	93	76:24
3	CuI (1.3 equiv)	0 °C, 2 h then RT, 2 h	83	86:14
4	CuI (10 mol%)	RT, 2 h	39	83:17
5	CuCN (10 mol%)	RT, 2 h	39	82:18
6[c]	$CuBr{\cdot}SMe_2$ (10 mol%)	0 °C, 2 h	87	92:8

[a] ^{19}F NMR yield using $PhCF_3$ as an internal standard. [b] The ratio of *E*-**12a** and **13a** was determined by ^{19}F NMR measurement. [c] **2** (1.2 equiv).

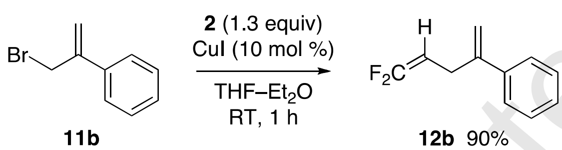
Other allylic bromides were difluorovinylated with **2** in the presence of Cu(I) catalysts. Allylic bromide *Z*-**11a**, a stereoisomer of *E*-**11a**, successfully reacted with **2** under the same conditions as those used in the reaction of *E*-**11a** to afford the diflurovinylated products *Z*-**12a**, *E*-**12a**, and **13a** in 86% total yield (*Z*-**12a**/*E*-**12a**/**13a** = 90:2:8, eq. 5). In this reaction, difluorovinylation mainly proceeded on the carbon α to the bromine substituent with retention of stereochemistry. Furthermore, difluorovinylation of allylic bromide **11b** was readily effected in the presence of 10 mol % of CuI to provide the corresponding 1,1-difluoro-1,4-diene **12b** in 90% yield (eq. 6).



(4)



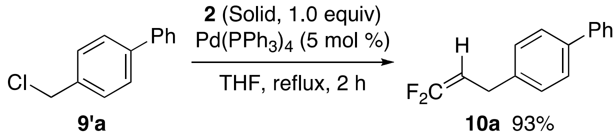
(5)



(6)

2.5. Use of the difluorovinylzinc–TMEDA complex as a powder

Removal of the solvents from the solution of the 2,2-difluorovinylzinc–TMEDA complex **2** under reduced pressure afforded a white powder of **2** containing LiCl. The solid-state **2** was found to be more thermally stable than **2** in solution. While **2** in solution was storable for a week at −20 °C under argon, solid-state **2** was unchanged after being stored for more than a month at 0 °C under argon. Furthermore, the reactivity of solid-state **2** was comparable to that of **2** in solution. On treatment with solid-state **2** in the presence of 5 mol % of $Pd(PPh_3)_4$, 4-phenylbenzyl chloride (**9'a**) successfully underwent difluorovinylation to produce 4-(3,3-difluoroallyl)biphenyl **10a** in 93% yield (eq. 7), whereas the reaction with **2** in solution afforded **10a** in 92% yield (Table 5, entry 1).


(7)

3. Conclusion

We have demonstrated a versatile method for accessing 2,2-difluorovinyl compounds via palladium- or copper-catalyzed cross-coupling of the difluorovinylzinc–TMEDA complex derived from 1,1-difluoroethylene, an industrial material. Difluorovinylation of alkenyl, alkynyl, allyl, and benzyl halides, as well as aryl halides, was thus successfully achieved. As a powder, the difluorovinylzinc–TMEDA complex is storable for a longer duration than its solution and can be used as an easily-handled difluorovinylation reagent.

4. Experimental

4.1. General

^{1}H NMR, ^{13}C NMR, and ^{19}F NMR spectra were recorded on a Bruker Avance 500. Chemical shift values are given in ppm relative to internal Me_4Si (for ^{1}H NMR: δ = 0.00 ppm), $CDCl_3$ (for ^{13}C NMR: δ = 77.0 ppm), and C_6F_6 (for ^{19}F NMR: δ = 0.00 ppm). IR spectra were recorded on a Horiba FT-300S spectrometer by the attenuated total reflectance (ATR) method. Mass spectra were measured on a JEOL JMS-T100GCV. Elemental analyses were carried out at the Elemental Analysis Laboratory, Division of Chemistry, Faculty of Pure and Applied Sciences, University of Tsukuba. All reactions were carried out under argon. Column chromatography was performed on silica gel (Kanto Chemical Co. Inc., Silica Gel 60). Tetrahydrofuran (THF) and diethyl ether were purified by a solvent-purification system (GlassContour) equipped with columns of activated alumina and supported-copper catalyst (Q-5) before use. *N*,*N*,*N′*,*N′*-Tetramethylethylenediamine (TMEDA) was distilled from KOH.

*4.2. Preparation of 2,2-difluorovinylzinc chloride–TMEDA (1/1) (**2**)*

To a solution of TMEDA (98 μL, 0.65 mmol) in THF (2.0 mL) and diethyl ether (0.50 mL) at −110 °C was slowly added gaseous 1,1-difluoroethylene (14.5 mL, 0.60 mmol) via syringe, and the mixture was stirred at the same temperature for 5 min. *sec*-BuLi (0.96 M in hexane, 0.52 mL, 0.50 mmol) was added dropwise to the solution at −110 °C, and then the mixture was stirred at the same temperature for 20 min. To the reaction mixture at −110 °C was added a THF solution of anhydrous $ZnCl_2$ (1.00 M, 0.50 mL, 0.50 mmol). After the reaction mixture was stirred at −100 °C for 30 min, a THF–diethyl ether solution of **2** was obtained as a colorless solution (0.48 mmol, 95%: The yield and the concentration were determined by ^{19}F NMR using $PhCF_3$ as an internal standard): ^{19}F NMR (470 MHz, C_6D_6) δ 87.9 (1F, dd, J_{FF} = 58 Hz, J_{FH} = 58 Hz), 98.7 (1F, dd, J_{FF} = 58 Hz, J_{FH} = 15 Hz).

*4.3. Synthesis of β,β-difluorostyrenes **4** by Pd-catalyzed cross-coupling of **2** with aryl halides **3***

*4.3.1. Typical procedure for the synthesis of β,β-difluorostyrenes **4***

To the solution of **2** (0.125 M in THF and diethyl ether, 7.6 mL, 0.95 mmol) were added a solution of 4-iodoanisole (**3b**, 189 mg, 0.81 mmol) in THF (1.5 mL) and $Pd(PPh_3)_4$ (17 mg, 15 μmol). After being refluxed for 6 h, the reaction mixture was filtered through a pad of silica gel (diethyl ether). The filtrate was concentrated under reduced pressure and purified by preparative thin layer chromatography on silica gel (pentane/diethyl ether = 20:1) to give **4b** (119 mg, 87%).

*4.3.2. Spectral data of β,β-difluorostyrenes **4***

*4.3.2.1. 1-(2,2-Difluorovinyl)-4-methylbenzene (**4a**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 2.33 (s, 3H), 5.23 (dd, J_{HF} = 26.4, 3.8 Hz, 1H), 7.14 (d, J = 8.0 Hz, 2H), 7.22 (d, J = 8.0 Hz, 2H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 21.1, 81.9 (dd, J_{CF} = 29, 14 Hz), 127.3–127.5 (2C, m), 129.3, 136.7, 156.0 (dd, J_{CF} = 298, 288 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 77.9 (dd, J_{FF} = 33 Hz, J_{FH} = 4 Hz, 1F), 79.9 (dd, J_{FF} = 33 Hz, J_{FH} = 26 Hz, 1F). The NMR spectral data described above showed good agreement with the literature data [10b].

*4.3.2.2. 1-(2,2-Difluorovinyl)-4-methoxybenzene (**4b**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 3.80 (s, 3H), 5.23 (dd, J_{HF} = 26.4, 3.8 Hz, 1H), 6.87 (d, J = 8.7 Hz, 2H), 7.25 (d, J = 8.7 Hz, 2H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 55.2, 81.5 (dd, J_{CF} = 29, 14 Hz), 114.1, 122.7 (dd, J_{CF} = 6, 6 Hz), 128.7 (dd, J_{CF} = 7, 4 Hz), 155.8 (dd, J_{CF} = 297, 287 Hz), 158.5. ^{19}F NMR (470 MHz, $CDCl_3$): δ 76.5 (dd, J_{FF} = 37 Hz, J_{FH} = 4 Hz, 1F), 78.3 (dd, J_{FF} = 37 Hz, J_{FH} = 26 Hz, 1F). The NMR spectral data described above showed good agreement with the literature data [15].

*4.3.2.3. 2-(2,2-Difluorovinyl)aniline (**4c**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 3.80 (s, 2H), 5.21 (dd, J_{HF} = 25.2, 2.9 Hz, 1H), 6.73 (d, J = 7.9 Hz, 1H), 6.80 (dd, J = 7.6, 7.5 Hz, 1H), 7.10 (dd, J = 7.9, 7.5, Hz, 1H), 7.24 (d, J = 7.6, 1H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 77.0 (dd, J_{CF} = 29, 16 Hz), 115.8, 116.2, 119.4, 128.6, 129.4 (dd, J_{CF} = 6, 2 Hz), 143.2, 156.6 (dd, J_{CF} = 297, 289 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 78.1 (dd, J_{FF} = 29 Hz, J_{FH} = 25 Hz, 1F), 79.3 (dd, J_{FF} = 29 Hz, J_{FH} = 3 Hz, 1F). IR (neat): 3384, 1732, 1236, 935, 771 cm^{-1}. HRMS (EI): m/z calcd. for $C_8H_7F_2N$ ($[M]^+$): 155.0547; Found: 155.0548.

*4.3.2.4. 1-(2,2-Difluorovinyl)-4-nitrobenzene (**4d**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 5.41 (dd, J_{HF} = 25.5, 3.3 Hz, 1H), 7.49 (d, J = 8.9 Hz, 2H), 8.21 (d, J = 8.9 Hz, 2H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 81.6 (dd, J_{CF} = 31, 13 Hz), 124.0, 128.1 (dd, J_{CF} = 7, 4 Hz), 137.3 (dd, J_{CF} = 7, 7 Hz), 146.4, 157.1 (dd, J_{CF} = 302, 293 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 83.4 (dd, J_{FF} = 18 Hz, J_{FH} = 3 Hz, 1F), 84.9 (dd, J_{FH} = 26 Hz, J_{FF} = 18 Hz, 1F). The NMR spectral data described above showed good agreement with the literature data [10b].

*4.3.2.5. 1-(2,2-Difluorovinyl)naphthalene (**4e**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 5.85 (dd, J_{HF} = 24.4, 3.3 Hz, 1H), 7.45 (dd, J = 7.7, 7.7 Hz, 1H), 7.47–7.53 (m, 2H), 7.57 (d, J = 7.2 Hz, 1H), 7.77 (d, J = 8.2 Hz, 1H), 7.83–7.85 (m, 1H), 7.93 (d, J = 8.2 Hz, 1H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 78.6 (dd, J_{CF} = 29, 16 Hz), 123.7, 125.4, 125.9, 126.3, 126.4–126.5 (overlapped dd), 126.5 (dd, J_{CF} = 7, 2 Hz), 127.9, 128.7, 131.4 (d, J_{CF} = 3 Hz), 133.6, 156.7 (dd, J_{CF} = 297, 289 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 77.8 (dd, J_{FF} = 29 Hz, J_{FH} = 24 Hz, 1F), 79.7 (dd, J_{FF} = 29 Hz, J_{FH} = 3 Hz, 1F). The NMR spectral data described above showed good agreement with the literature data [16].

*4.3.2.6. 2-(2,2-Difluorovinyl)naphthalene (**4f**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 5.39 (dd, J_{HF} = 26.2, 3.6 Hz, 1H), 7.41–7.47 (m, 3H), 7.71 (s, 1H), 7.73–7.79 (m, 3H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 82.4 (dd, J_{CF} = 29, 14 Hz), 125.3 (dd, J_{CF} = 7, 2 Hz), 126.0, 126.4, 126.6 (dd, J_{CF} = 6, 6 Hz), 127.6, 127.7, 127.8 (dd, J_{CF} = 7, 7 Hz), 128.3, 132.2, 133.4, 156.4 (dd, J_{CF} = 299, 289 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 79.2 (dd, J_{FF} = 31 Hz, J_{FH} = 4 Hz, 1F), 80.9 (dd, J_{FF} = 31 Hz, J_{FH} = 26 Hz, 1F). The NMR spectral data described above showed good agreement with the literature data [9a].

*4.3.2.7. 2-(2,2-Difluorovinyl)biphenyl (**4g**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 5.25 (dd, J_{HF} = 26.1, 4.0 Hz, 1H), 7.33–7.37 (m, 4H), 7.38–7.41 (m, 2H), 7.44–7.47 (m, 2H), 7.64 (d, J = 7.4 Hz, 1H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 80.6 (dd, J_{CF} = 30, 13 Hz), 127.1, 127.3, 127.5, 127.9 (d, J_{CF} = 6 Hz), 128.1 (d, J_{CF} = 10 Hz), 128.3, 129.5, 130.1, 140.7, 141.2 (d, J_{CF} = 4 Hz), 156.2 (dd, J_{CF} = 288 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 78.1 (dd, J_{FF} = 32 Hz, J_{FH} = 26 Hz, 1F), 79.9 (dd, J_{FF} = 32 Hz, J_{FH} = 4 Hz, 1F). The NMR spectral data described above showed good agreement with the literature data [17].

*4.3.2.8. 2-[2-(2,2-Difluorovinyl)phenyl]-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (**4h**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 1.35 (s, 12H), 6.25 (dd, J_{HF} = 26.8, 5.1 Hz, 1H), 7.22 (ddd, J = 7.6, 7.4, 1.0 Hz, 1H), 7.36–7.47 (m, 1H), 7.55 (dd, J = 8.0, 1.0 Hz, 1H), 7.84 (dd, J = 7.6, 1.0 Hz, 1H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 24.8, 82.4 (dd, J_{CF} = 30, 11 Hz), 83.8, 126.1, 127.5 (d, J_{CF} = 10 Hz), 131.2, 136.2 (dd, J_{CF} = 7, 6 Hz), 136.4, 156.2 (dd, J_{CF} = 299, 287 Hz), (One aromatic carbon signal was not detected due to ^{13}C–^{10}B and ^{13}C–^{11}B coupling and overlapping with other signals). ^{19}F NMR (470 MHz, $CDCl_3$): δ 79.2 (dd, J_{FF} 32 Hz, J_{FH} = 5 Hz, 1F), 79.6 (dd, J_{FF} = 32 Hz, J_{FH} = 27 Hz, 1F). IR (neat): 2979, 1724, 1346, 1146, 912, 743 cm^{-1}. HRMS (EI): m/z calcd. for $C_{14}H_{17}BF_2O_2$ ($[M]^+$): 266.1290; Found: 266.1288.

*4.3.2.9. 2-(2,2-Difluorovinyl)-1,3-dimethoxybenzene (**4i**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 3.83 (s, 6H), 5.18 (dd, J_{HF} = 27.7, 2.6 Hz, 1H), 6.56 (d, J = 8.4 Hz, 2H), 7.22 (t, J = 8.4 Hz, 1H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 55.8, 72.1 (dd, J_{CF} = 34, 19 Hz), 103.6, 107.3 (dd, J_{CF} = 7, 4 Hz), 128.9, 155.4 (dd, J_{CF} = 297, 285 Hz), 157.9. ^{19}F NMR (470 MHz, $CDCl_3$): δ 77.9

(dd, J_{FF} = 26 Hz, J_{FH} = 3 Hz, 1F), 84.4 (dd, J_{FH} = 28 Hz, J_{FF} = 26 Hz, 1F). IR (neat): 2941, 1738, 1473, 1254, 1109 cm^{-1}. Anal. calcd. for $C_{10}H_{10}F_2O_2$: C, 60.00; H, 5.04; Found: C, 59.92; H, 5.09.

*4.3.2.10. 3-(2,2-Difluorovinyl)phenyl trifluoromethanesulfonate (**4j**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 5.32 (dd, J_{HF} = 25.3, 3.3 Hz, 1H), 7.16 (dd, J = 8.2, 2.3 Hz, 1H), 7.25 (dd, J = 2.3, 1.5 Hz, 1H), 7.34 (d, J = 8.0, 1H), 7.42 (dd, J = 8.2, 8.0 Hz, 1H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 81.3 (dd, J_{CF} = 31, 13 Hz), 118.7 (q, J_{CF} = 321 Hz), 119.7, 120.2 (dd, J_{CF} = 7, 3 Hz), 127.5 (dd, J_{CF} = 6, 4 Hz), 130.4, 133.1 (dd, J_{CF} = 7, 7 Hz), 149.8, 156.8 (dd, J_{CF} = 300, 292 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 81.6 (dd, J_{FF} = 25 Hz, J_{FH} = 3 Hz, 1F), 83.4 (dd, J_{FF} = 25 Hz, J_{FH} = 25 Hz, 1F), 90.0 (s, 3F). IR (neat): 1728, 1423, 1213, 1140, 906, 845, 771 cm^{-1}. HRMS (EI): *m/z* calcd. for $C_9H_5F_5O_3S$ ($[M]^+$): 287.9880; Found: 287.9879.

*4.3.2.11. 3-Bromo-4-(2,2-difluorovinyl)biphenyl (**4k**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 5.72 (dd, J_{HF} = 25.6, 3.6 Hz, 1H), 7.37 (tt, J = 7.4, 1.3 Hz, 1H), 7.43–7.47 (m, 2H), 7.50–7.62 (m, 4H), 7.82 (d, J = 1.9 Hz, 1H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 81.4 (dd, J_{CF} = 33, 12 Hz), 123.6 (dd, J_{CF} = 6, 2 Hz), 126.2, 126.9, 128.0, 128.9, 129.16, 129.23, 131.3, 139.0, 141.6, 156.7 (dd, J_{CF} = 299, 290 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 77.7 (dd, J_{FF} = 30 Hz, J_{FH} = 26 Hz, 1F), 79.6 (dd, J_{FF} = 30 Hz, J_{FH} = 4 Hz, 1F). IR (neat): 1726, 1477, 1248, 1178, 945, 758 cm^{-1}. HRMS (EI): *m/z* calcd. for $C_{14}H_9BrF_2$ ($[M]^+$): 293.9856; Found: 293.9849.

*4.3.2.12. 3,5-Dichloro-4-(2,2-difluorovinyl)biphenyl (**4l**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 5.34 (d, J_{HF} = 25.9 Hz, 1H), 7.39 (t, J = 7.5 Hz, 1H), 7.44 (dd, J = 7.5, 7.1 Hz, 2H), 7.52 (d, J = 7.1 Hz, 2H), 7.55 (s, 2H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 76.6–76.9 (overlapped dd), 126.2 (dd, J = 9, 4 Hz), 126.5, 126.9, 128.6, 129.1, 135.9, 138.0, 142.9, 155.8 (dd, J_{CF} = 297, 290 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 79.6 (d, J_{FF} = 20 Hz, 1F), 85.9 (dd, J_{FH} = 26, J_{FF} = 20 Hz, 1F). IR (neat): 1738, 1535, 1248, 1173, 914, 758 cm^{-1}. HRMS (EI): *m/z* calcd. for $C_{14}H_8Cl_2F_2$ ($[M]^+$): 283.9971; Found: 283.9964.

*4.3.2.13. 3'-Chloro-2-(2,2-difluorovinyl)biphenyl (**4m**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 5.17 (dd, J_{HF} = 25.8, 4.1 Hz, 1H), 7.17–7.22 (m, 1H), 7.25–7.40 (m, 6H), 7.60 (d, J = 7.8 Hz, 1H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 80.3 (dd, J_{CF} = 30, 13 Hz), 127.2, 127.5, 127.8, 127.9 (dd, J_{CF} = 7, 6 Hz), 128.0, 128.2 (dd, J_{CF} = 9, 1 Hz), 129.49, 129.51, 130.0, 134.2, 139.7 (d, J_{CF} = 3.5 Hz), 142.5, 156.3 (dd, J_{CF} = 299, 288 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 77.7 (dd, J_{FF} = 30 Hz, J_{FH} = 26 Hz, 1F), 79.6 (dd, J_{FF} = 30 Hz, J_{FH} = 4 Hz, 1F). IR (neat): 3062, 1724, 1232, 1173, 939, 756 cm^{-1}. Anal. calcd. for $C_{14}H_9ClF_2$: C, 67.08; H, 3.62; Found: C, 67.12; H, 3.72.

*4.3.2.14. 1-Bromo-3-(2,2-difluorovinyl)benzene (**4n**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 5.23 (dd, J_{HF} = 25.8, 3.5 Hz, 1H), 7.20 (dd, J = 7.8, 7.8 Hz, 1H), 7.22–7.27 (m, 1H), 7.37 (d, J = 7.8 Hz, 1H), 7.48 (s, 1H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 81.3 (dd, J_{CF} = 30, 14 Hz), 122.7, 126.1 (dd, J_{CF} = 6, 4 Hz), 130.0, 130.1, 130.4 (dd, J_{CF} = 7, 4 Hz), 132.4 (dd, J_{CF} = 7, 7 Hz), 156.5 (dd, J_{CF} = 300, 255 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 80.4 (dd, J_{FF} = 27, J_{FH} = 3 Hz, 1F), 82.4 (dd, J_{FF} = 27 Hz, J_{FH} = 26 Hz, 1F). The NMR spectral data described above showed good agreement with the literature data [6c].

*4.4. Synthesis of 1,1-difluoro-1,3-dienes **6** by Pd-catalyzed cross-coupling of **2** with alkenyl halides **5***

*4.4.1. Typical procedure for the synthesis of 1,1-difluoro-1,3-dienes **6***

To the solution of **2** (0.11 M in THF and diethyl ether, 2.5 mL, 0.27 mmol) were added a solution of (*E*)-1-(2-bromovinyl)-4-(trifluoromethyl)benzene (**5c**, 68 mg, 0.27 mmol) in THF (0.5 mL) and $Pd(PPh_3)_4$ (6 mg, 5 μmol). After being refluxed refluxing for 2 h, the reaction mixture was filtered through a pad of silica gel (diethyl ether). The filtrate was concentrated under reduced pressure and purified by preparative thin layer chromatography on silica gel (pentane) to give **6c** (55 mg, 86%).

*4.4.2. Spectral data of 1,1-difluoro-1,3-dienes **6***

*4.4.2.1. (E)-1-(4,4-Difluorobuta-1,3-dienyl)-4-methylbenzene (**6a**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 2.33 (s, 3H), 5.10 (dddd, J_{HF} = 24.6 Hz, J = 10.9 Hz, 1H), 6.43 (d, J = 15.9 Hz, 1H), 6.60 (dd, J = 15.9, 10.9 Hz, 1H), 7.11 (d, J = 8.0 Hz, 2H), 7.27 (d, J = 8.0 Hz, 2H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 21.2, 82.9 (dd, J_{CF} = 30, 17 Hz), 116.8 (dd, J_{CF} = 4, 2 Hz), 126.0, 129.3, 131.0 (dd, J_{CF} = 13, 4 Hz), 134.1, 137.5, 156.7 (dd, J_{CF} = 321, 314 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 75.3 (d, J_{FF} = 28 Hz, 1F), 77.1 (dd, J_{FF} = 28 Hz, J_{FH} = 24 Hz, 1F). IR (neat): 2924, 2854, 1747, 1716, 1512, 1456, 1248, 1180, 1142, 796, 748 cm^{-1}. HRMS (EI): m/z calcd. for $C_{11}H_{10}F_2$ ([M]$^+$): 180.0751; Found: 180.0748.

*4.4.2.3. (E)-1-(4,4-Difluorobuta-1,3-dienyl)-4-methoxybenzene (**6b**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 3.81 (s, 3H), 5.10 (ddd, J_{HF} = 24.3 Hz, J = 10.6 Hz, J_{HF} = 1.5 Hz, 1H), 6.42 (d, J = 15.9 Hz, 1H), 6.51 (dd, J = 15.9, 10.6 Hz, 1H), 6.85 (d, J = 8.8 Hz, 2H), 7.32 (d, J = 8.8 Hz, 2H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 55.3, 82.9 (dd, J_{CF} = 30, 18 Hz), 114.1, 115.7 (d, J_{CF} = 4 Hz), 127.4, 129.8, 130.6 (dd, J_{CF} = 11, 3 Hz), 156.6 (dd, J_{CF} = 297, 292 Hz), 159.3. ^{19}F NMR (470 MHz, $CDCl_3$): δ 74.8 (d, J_{FF} = 29 Hz, 1F), 76.6 (dd, J_{FF} = 29 Hz, J_{FH} = 24 Hz, 1F). IR (neat): 2923, 2852, 1716, 1606, 1510, 1458, 1377, 1254, 1178, 1124, 1034, 910, 737 cm^{-1}. HRMS (EI): *m*/*z* calcd. for $C_{11}H_{10}F_2O$ ($[M]^+$): 196.0700; Found: 196.0703.

*4.4.2.4. (E)-1-(4,4-Difluorobuta-1,3-dienyl)-4-(trifluoromethyl)benzene (**6c**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 5.17 (dd, J_{HF} = 23.8 Hz, *J* = 11.0 Hz, 1H), 6.50 (d, *J* = 15.9 Hz, 1H), 6.75 (dd, *J* = 15.9, 11.0 Hz, 1H), 7.47 (d, *J* = 8.3 Hz, 2H), 7.56 (d, *J* = 8.3 Hz, 2H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 82.7 (dd, J_{CF} = 28, 17 Hz), 120.4 (dd, J_{CF} = 4, 2 Hz), 124.1 (q, J_{CF} = 272 Hz), 125.6 (q, J_{CF} = 4 Hz), 126.2, 129.3 (q, J_{CF} = 33 Hz), 129.5 (dd, J_{CF} = 12, 3 Hz), 140.3, 157.2 (dd, J_{CF} = 299, 293 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 77.7 (d, J_{FF} = 23 Hz, 1F), 79.2 (dd, J_{FH} = 24 Hz, J_{FF} = 23 Hz, 1F), 100.3 (s, 3F). IR (neat): 1714, 1616, 1323, 1281, 1167, 1124, 1068, 937, 810 cm^{-1}. HRMS (EI): *m*/*z* calcd. for $C_{11}H_7F_5$ ($[M]^+$): 234.0468; Found: 234.0466.

*4.4.2.5. [4-(2,2-Difluorovinyl)-6,6-difluorohexa-3,3-dienyl] benzene (**6d**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 2.39 (td, *J* = 7.6, 7.5 Hz, 2H), 2.69 (t, *J* = 7.6 Hz, 2H), 4.87 (dd, J_{HF} = 24.4 Hz, 2H), 5.60 (t, *J* = 7.5 Hz, 1H), 7.16–7.22 (m, 3H), 7.28 (dd, *J* = 7.6, 7.6 Hz, 2H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 30.7, 35.1, 77.2 (dd, J_{CF} = 29, 16 Hz), 82.3 (dd, J_{CF} = 29, 13 Hz), 119.2 (dd, J_{CF} = 9, 5 Hz), 126.1, 128.40, 128.40, 133.7 (dddd, J_{CF} = 17, 11, 6, 2 Hz), 141.2, 155.7 (dd, J_{CF} = 297, 287), 155.8 (dd, J_{CF} = 297, 288). ^{19}F NMR (470 MHz, $CDCl_3$): δ 75.5 (dd, J_{FF} = 34 Hz, J_{FH} = 4 Hz, 1F), 76.76 (ddd, J_{FF} = 34 Hz, J_{FH} = 24, 3 Hz, 1F), 76.77 (d, J_{FF} = 30 Hz, 1F), 79.9 (ddd, J_{FF} = 30 Hz, J_{FH} = 24, 4 Hz, 1F).

IR (neat): 2927, 2850, 1541, 1508, 1219, 771 cm^{-1}. HRMS (EI): *m*/*z* calcd. for $C_{14}H_{12}F_4$ ($[M]^+$): 256.0875; Found: 256.0876.

*4.5. Synthesis of 1,1-difluoro-1,3-enynes **8** by Pd-catalyzed cross-coupling of **2** with alkynyl halides 7*

*4.5.1. Typical procedure for the synthesis of 1,1-difluoro-1,3-enynes **8***

In a two-necked flask were placed $Pd_2(dba)_3 \cdot CHCl_3$ (8 mg, 8 μmol), dppp (6 mg, 0.02 mmol), and **2** (0.12 M in THF and diethyl ether, 3.2 mL, 0.38 mmol). After stirring for 10 min, 2-(4-iodobut-3-ynyl)naphthalene (**7b**, 77 mg, 0.30 mmol) was added to the mixture. After refluxing for 2 h, the reaction mixture was filtered through a pad of silica gel (diethyl ether). The filtrate was concentrated under reduced pressure and purified by silica gel column chromatography (hexane) to give **8b** (68 mg, 94%).

*4.5.2. Spectral data of 1,1-difluoro-1,3-enynes **8***

*4.5.2.1. (6,6-Difluorohex-5-en-3-ynyl)benzene (**8a**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 2.58 (tdd, J = 7.5, 1.7 Hz, J_{HF} = 1.1 Hz, 2H), 2.84 (t, J = 7.5 Hz, 2H), 4.52 (dtd, J_{HF} = 23.3 Hz, J = 1.7 Hz, J_{HF} = 0.6 Hz, 1H), 7.20–7.22 (m, 3H), 7.29 (dd, J = 7.5, 7.5 Hz, 2H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 21.6, 34.9, 65.4 (dd, J_{CF} = 42, 19 Hz), 69.2 (dd, J_{CF} = 13, 3 Hz), 93.5 (dd, J_{CF} = 9, 4 Hz), 126.3, 128.38, 128.43, 140.4, 162.0 (dd, J_{CF} = 300, 293 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 81.1 (d, J_{FF} = 10 Hz, 1F), 86.3 (dd, J_{FH} = 23 Hz, J_{FF} = 10 Hz, 1F). IR (neat): 2956, 2931, 1722, 1346, 1238, 914, 773, 698 cm^{-1}. HRMS (EI): *m*/*z* calcd. for $C_{12}H_{10}F_2$ ($[M]^+$): 192.0751; Found: 192.0749.

*4.5.2.2. 2-(6,6-Difluorohex-5-en-3-ynyl)naphthalene (**8b**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 2.68 (t, J = 7.5 Hz, 2H), 3.01 (t, J = 7.5 Hz, 2H), 4.53 (d, J_{HF} = 23.4 Hz, 1H), 7.35 (dd, J = 8.4, 1.8 Hz, 1H), 7.43 (ddd, J = 8.4, 6.9, 1.1 Hz, 1H), 7.46 (ddd, J = 8.4, 6.9, 1.1 Hz, 1H), 7.67 (s, 1H), 7.77–7.82 (m, 3H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 21.6, 35.0, 65.4 (dd, J_{CF} = 42, 19 Hz), 69.4 (dd, J_{CF} = 12, 5 Hz), 93.5 (dd, J_{CF} = 9, 4 Hz), 125.4, 126.0, 126.7, 127.1, 127.5, 127.6, 128.0, 132.2, 133.5, 137.9, 162.0 (dd, J_{CF} = 299, 293 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 80.0 (d, J_{FF} = 10 Hz, 1F), 85.2 (dd, J_{FH} = 23 Hz, J_{FF} = 10 Hz, 1F). IR (neat): 3055, 1720, 1508, 1344, 1234, 1165, 1124, 928, 910, 814, 744 cm^{-1}. HRMS (EI): m/z calcd. for $C_{16}H_{12}F_2$ ($[M]^+$): 242.0907; Found: 242.0911.

*4.5.2.3. 2-(4,4-Difluorobut-3-en-1-ynyl)biphenyl **(8c)***

^{1}H NMR (500 MHz, $CDCl_3$): δ 4.67 (d, J_{HF} = 23.3 Hz, 1H), 7.27–7.31 (m, 1H), 7.34–7.43 (m, 5H), 7.54–7.58 (m, 3H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 65.8 (dd, J_{CF} = 42, 19 Hz), 80.4 (dd, J_{CF} = 12, 5 Hz), 92.9 (dd, J_{CF} = 9, 4 Hz), 121.1, 127.0, 127.5, 127.9, 128.7, 129.1, 129.5, 132.9, 140.2, 143.6, 161.7 (dd, J_{CF} = 302, 295 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 83.5 (d, J_{FF} = 3 Hz, 1F), 88.9 (dd, J_{FH} = 23 Hz, J_{FF} = 3 Hz, 1F). IR (neat): 3053, 2960, 2920, 1728, 1281, 1173, 958, 827, 767 cm^{-1}. HRMS (EI): m/z calcd. for $C_{16}H_{10}F_2$ ($[M]^+$): 240.0751; Found: 240.0746.

*4.5.2.4. 1-(4,4-Difluorobut-3-en-1-ynyl)-4-methoxybenzene **(8d)***

^{1}H NMR (500 MHz, $CDCl_3$): δ 3.81 (s, 3H), 4.78 (d, J_{HF} = 23.1 Hz, 1H), 6.84 (d, J = 8.8 Hz, 2H), 7.37 (d, J = 8.8 Hz, 2H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 55.3, 65.8 (dd, J_{CF} = 42, 19 Hz), 76.1 (dd, J_{CF} = 12, 5 Hz), 93.1 (dd, J_{CF} = 8, 4 Hz), 114.0, 114.9, 132.9, 159.8, 161.6 (dd, J_{CF} = 300, 295 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 81.1 (d, J_{FF} = 6 Hz, 1F), 86.5 (dd, J_{FH} = 23 Hz, J_{FF} = 6 Hz, 1F). IR (neat): 2960, 2837, 1714, 1604, 1508, 1464, 1348, 1282, 1246, 1207, 1170, 1051, 1030, 908, 831, 771 cm^{-1}. HRMS (EI): *m*/*z* calcd. for $C_{11}H_8F_2O$ ($[M]^+$): 194.0543; Found: 194.0547.

*4.5.2.5. 1-(4,4-Difluorobut-3-en-1-ynyl)-4-nitrobenzene (**8e**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 4.86 (dd, J_{HF} = 22.9, 1.0 Hz, 1H), 6.84 (d, J = 8.8 Hz, 2H), 8.19 (d, J = 8.8 Hz, 2H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 65.4 (dd, J_{CF} = 43, 19 Hz), 83.0 (dd, J_{CF} = 12, 5 Hz), 91.4 (dd, J_{CF} = 9, 4 Hz), 123.6, 129.6, 132.0, 147.1, 162.2 (dd, J_{CF} = 303, 297 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 85.1 (d, J_{FF} = 2 Hz, 1F), 90.0 (dd, J_{FH} = 23 Hz, J_{FF} = 2 Hz, 1F). IR (neat): 1716, 1593, 1522, 1344, 1296, 1201, 914, 854, 744 cm^{-1}. HRMS (EI): *m*/*z* calcd. for $C_{10}H_5F_2NO_2$ ($[M]^+$): 209.0288; Found: 209.0294.

*4.6. Synthesis of (3,3-difluoroallyl)arenes **10** by Pd-catalyzed cross-coupling of **2** with benzyl halides **9'***

*4.6.1. Typical procedure for the synthesis of (3,3-difluoroallyl)arenes **10***

In a two-necked flask was placed **2** (0.11 M in THF and diethyl ether, 9.1 mL, 1.0 mmol). To the solution were added a solution of 1-butyl-4-(chloromethyl)benzene (**9'b**, 146 mg, 0.80 mmol) in THF (0.5 mL) and $Pd(PPh_3)_4$ (46 mg, 4.0 μmol). After refluxing for 2 h, the reaction mixture was filtered through a pad of silica gel (diethyl ether). The filtrate was concentrated under reduced pressure and purified by preparative thin layer chromatography on silica gel (pentane) to give **10b** (157 mg, 93%).

*4.6.2. Spectral data of (3,3-difluoroallyl)arenes **10***

*4.6.2.1. 4-(3,3-Difluoroallyl)biphenyl (**10a**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 3.37 (d, J = 8.1 Hz, 2H), 4.43 (dtd, J_{HF} = 24.8 Hz, J = 8.1 Hz, J_{HF} = 2.2 Hz, 1H), 7.27 (d, J = 8.2 Hz, 2H), 7.34 (tt, J = 7.5, 1.3 Hz, 1H), 7.43 (dd, J = 7.5, 7.5 Hz, 2H), 7.53 (d, J = 8.2 Hz, 2H), 7.57 (dd, J = 7.5, 1.3 Hz, 2H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 28.0 (d, J_{CF} = 5 Hz), 77.6 (dd, J_{CF} = 23, 20 Hz), 127.0, 127.2, 127.3, 128.5, 128.7, 138.5 (dd, J_{CF} = 2, 2 Hz), 139.5, 140.9, 156.6 (dd, J_{CF} = 289, 288 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 71.3 (dd, J_{FF} = 45 Hz, J_{FH} = 25 Hz, 1F), 74.2 (d, J_{FF} = 45 Hz, 1F). IR (neat): 3030, 1747, 1489, 1294, 1230, 1173, 964, 758, 694 cm^{-1}. HRMS (EI): m/z calcd. for $C_{15}H_{12}F_2$ ($[M]^+$): 230.0907; Found: 230.0904.

*4.6.2.2. 1-Butyl-4-(3,3-difluoroallyl)benzene **(10b)***

^{1}H NMR (500 MHz, $CDCl_3$): δ 0.92 (t, J = 7.4 Hz, 3H), 1.34 (qt, J = 7.4, 7.4 Hz, 2H), 1.58 (tt, J = 7.8, 7.4 Hz, 2H), 2.57 (t, J = 7.8 Hz, 2H), 3.27 (d, J = 8.0 Hz, 2H), 4.36 (dtd, J_{HF} = 24.8 Hz, J = 8.0 Hz, J_{HF} = 2.3 Hz, 1H), 7.08 (d, J = 8.2 Hz, 2H), 7.11 (d, J = 8.2 Hz, 2H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 13.9, 22.4, 28.0 (d, J_{CF} = 5 Hz), 33.7, 35.2, 77.8 (dd, J_{CF} = 23, 20 Hz), 127.9, 128.6, 136.6 (d, J_{CF} = 2 Hz), 141.1, 156.6 (dd, J_{CF} = 288, 286 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 69.9 (dd, J_{FF} = 46 Hz, J_{FH} = 25 Hz, 1F), 72.8 (d, J_{FF} = 46 Hz, 1F). IR (neat): 2958, 2929, 2858, 1745, 1514, 1344, 1288, 1230, 1171, 958, 802, 758 cm^{-1}. HRMS (EI): m/z calcd. for $C_{13}H_{16}F_2$ ($[M]^+$): 210.1220; Found: 210.1222.

*4.6.2.3. 1-(3,3-Difluoroallyl)-4-methoxybenzene **(10c)***

^{1}H NMR (500 MHz, $CDCl_3$): δ 3.19 (d, J = 8.0 Hz, 2H), 3.71 (s, 3H), 4.28 (dtd, J_{HF} = 24.9 Hz, J = 8.0 Hz, J_{HF} = 2.3 Hz, 1H), 6.77 (d, J = 8.6 Hz, 2H), 7.03 (d, J = 8.6 Hz, 2H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 27.5 (d, J_{CF} = 5 Hz), 55.3, 78.0 (dd, J_{CF} = 23, 20 Hz), 114.0, 129.0, 131.5, 156.5 (dd, J_{CF} = 288, 287 Hz), 158.2. ^{19}F NMR (470 MHz, $CDCl_3$): δ 69.0 (dd, J_{FF} = 49 Hz, J_{FH} = 25 Hz, 1F), 71.9 (d, J_{FF} = 49 Hz, 1F). IR (neat): 2952, 1747, 1558, 1541, 1514, 1250, 1176, 1036, 741 cm^{-1}. HRMS (EI): *m*/*z* calcd. for $C_{10}H_{10}F_2O$ ($[M]^+$): 184.0700; Found: 184.0698.

*4.6.2.4. 1-(3,3-Difluoroallyl)-4-(trifluoromethyl)benzene (**10d**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 3.39 (d, J = 8.1 Hz, 2H), 4.39 (dtd, J_{HF} = 24.5 Hz, J = 8.1 Hz, J_{HF} = 2.1 Hz, 1H), 7.31 (d, J = 8.0 Hz, 2H), 7.56 (d, J = 8.0 Hz, 2H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 28.3 (d, J_{CF} = 5 Hz), 76.9 (dd, J_{CF} = 24, 20 Hz), 124.2 (q, J_{CF} = 272 Hz), 125.5 (q, J_{CF} = 4 Hz), 128.4, 128.9 (q, J_{CF} = 32 Hz), 143.5, 156.8 (dd, J_{CF} = 289, 288 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 72.1 (dd, J_{FF} = 43 Hz, J_{FH} = 25 Hz, 1F), 75.0 (dd, J_{FF} = 43 Hz, J_{FH} = 2 Hz, 1F), 100.4 (s, 3F). IR (neat): 2924, 2854, 1743, 1714, 1541, 1508, 1458, 1325, 1128, 1068, 760 cm^{-1}. HRMS (EI): *m*/*z* calcd. for $C_{10}H_7F_5$ ($[M]^+$): 222.0468; Found: 222.0461.

*4.7. Synthesis of 1,1-difluoro-1,4-dienes **12** by Cu-catalyzed cross-coupling of **2** with allyl halides **11***

*4.7.1. Typical procedure for the synthesis of 1,1-difluoro-1,4-dienes **12***

To the solution of **2** (0.10 M in THF and diethyl ether, 6.0 mL, 0.60 mmol) was added $CuBr{\cdot}SMe_2$ (10 mg, 50 μmol) at 0 °C. After being stirred for 5 min at the same temperature, a solution of (*E*)-(3-bromoprop-1-enyl)benzene (*E*-**11a**, 99 mg, 0.50 mmol) in THF (0.5 mL) was added. After being stirred for 2 h at 0 °C, the reaction mixture was filtered through a pad of silica gel (diethyl ether). The filtrate was concentrated under reduced pressure and purified by preparative thin layer chromatography on silica gel (pentane) to give **12a** (77 mg, 86%, *E*/*Z* = 92:8).

*4.7.2. Spectral data of 1,1-difluoro-1,4-dienes **12***

*4.7.2.1. (E)-(5,5-Difluoropenta-1,4-dienyl)benzene (E-**12a**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 2.90–2.94 (m, 2H), 4.31 (dtd, J_{HF} = 25.1 Hz, J = 7.9 Hz, J_{HF} = 2.3 Hz, 1H), 6.19 (dt, J = 15.8, 6.4 Hz, 1H), 6.46 (d, J = 15.8 Hz, 1H), 7.25 (t, J = 7.3 Hz, 1H), 7.34 (dd, J = 7.8, 7.3 Hz, 2H), 7.38 (d, J = 7.8 Hz, 2H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 25.6 (d, J_{CF} = 5 Hz), 76.3 (dd, J_{CF} = 22, 20 Hz), 126.1, 127.1, 127.2, 128.5, 130.7, 137.2, 156.5 (dd, J_{CF} = 288, 286 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 71.8 (dd, J_{FF} = 45 Hz, J_{FH} = 25 Hz, 1F), 74.5 (d, J_{FF} = 45 Hz, 1F). IR (neat): 3030, 2925, 2854, 1745, 1720, 1496, 1454, 1346, 1290, 1232, 1173, 958, 912, 742, 696 cm^{-1}. HRMS (EI): *m*/*z* calcd. for $C_{11}H_{10}F_2$ ($[M]^+$): 180.0751; Found: 180.0745.

*4.7.2.2. (Z)-(5,5-Difluoropenta-1,4-dienyl)benzene (Z-**12a**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 2.96–3.00 (m, 2H), 4.22 (dtd, J_{HF} = 25.2 Hz, J = 7.7 Hz, J_{HF} = 2.3 Hz, 1H), 5.59 (dt, J = 11.5, 7.4 Hz, 1H), 6.49 (d, J = 11.5 Hz, 1H), 7.23–7.26 (m, 3H), 7.34 (dd, J = 7.6, 7.6 Hz, 2H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 21.7 (d, J_{CF} = 5 Hz), 77.1 (dd, J_{CF} = 23, 20 Hz), 126.9, 128.3, 128.7, 129.0 (dd, J_{CF} = 2, 2 Hz), 130.2, 136.9, 156.4 (dd, J_{CF} = 288, 286 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 72.2 (dd, J_{FF} = 46 Hz, J_{FH} = 25 Hz, 1F), 73.9 (dd, J_{FF} = 46 Hz, J_{FH} = 2 Hz, 1F). IR (neat): 3022, 1741, 1495, 1446, 1338, 1284, 1230, 1174, 945, 914, 806, 698 cm^{-1}. HRMS (EI): *m*/*z* calcd. for $C_{11}H_{10}F_2$ ($[M]^+$): 180.0751; Found: 180.0750.

*4.7.2.3. (5,5-Difluoropenta-1,4-dien-2-yl)benzene (**12b**)*

^{1}H NMR (500 MHz, $CDCl_3$): δ 3.17 (d, J = 7.8 Hz, 2H), 4.27 (dtd, J_{HF} = 25.0 Hz, J = 7.8 Hz, J_{HF} = 2.3 Hz, 1H), 5.12 (d, J = 1.1 Hz, 1H), 5.36 (d, J = 1.1 Hz, 1H), 7.28 (tt, J = 7.3, 1.4 Hz, 1H), 7.32–7.35 (m, 2H), 7.39–7.41 (m, 2H). ^{13}C NMR (126 MHz, $CDCl_3$): δ 28.1 (d, J_{CF} = 5 Hz), 76.4 (dd, J_{CF} = 23, 20 Hz), 113.0, 125.9, 127.7, 128.4, 140.3, 145.6, 156.5 (dd, J_{CF} = 289, 286 Hz). ^{19}F NMR (470 MHz, $CDCl_3$): δ 71.3 (dd, J_{FF} = 48 Hz, J_{FH} = 25 Hz, 1F), 74.1 (dd, J_{FF} = 48 Hz, J_{FH} = 2 Hz, 1F). IR (neat): 2958, 2927, 1749, 1541, 1257, 1215, 769 cm^{-1}. HRMS (EI): *m/z* calcd. for $C_{11}H_{10}F_2$ ($[M]^+$): 180.0751; Found: 180.0752.

Acknowledgements

This research is partly supported by JSPS KAKENHI (Grant Number 25288016) and Asahi Glass Foundation. We acknowledge Central Glass Co., Ltd. for the generous gift of 1,1-difluoroethylene.

References

[1] (a) K. Uneyama, Organofluorine Chemistry, Blackwell Publishing, Oxford, UK, 2006.
(b) J. Ichikawa, Chim. Oggi 25(4) (2007) 54–57.
(c) H. Amii, K. Uneyama, Chem. Rev. 109 (2009) 2119–2183.
(d) T. Ahrens, J. Kohlmann, M. Ahrens, T. Braun, Chem. Rev. doi:10.1021/cr500257c.

[2] R. Souzy, B. Ameduri, B. Boutevin, Prog. Polym. Sci. 29 (2004) 75–106.

[3] M. Bobek, I. Kavai, E. De Clercq, J. Med. Chem. 30 (1987) 1494–1497.

[4] W.R. Moore, G.L. Schatzman, E.T. Jarvi, R.S. Gross, J.R. McCarthy, J. Am. Chem. Soc. 114 (1992) 360–361.

[5] (a) G. Magueur, B. Crousse, M. Ourévitch, D. Bonnet-Delpon, J.-P. Bégué, J. Fluorine Chem. 127 (2006) 637–642.
(b) J. Ichikawa, J. Synth. Org. Chem. Jpn. 68 (2010) 1175–1184.
(c) N.A. Meanwell, J. Med. Chem. 54 (2011) 2529–2591.

[6] For recent reports on the Wittig-type difluoromethylenation, see:
(a) J. Zheng, J. Cai, J.-H. Lin, Y. Guo, J.-C. Xiao, Chem. Commun. 49 (2013) 7513–7515.
(b) R. Loska, K. Szachowicz, D. Szydlik, Org. Lett. 15 (2013) 5706–5709.
(c) J. Zheng, J.-H. Lin, J. Cai, J.-C. Xiao, Chem. Eur. J. 19 (2013) 15261–15266.
(d) Q. Li, J.-H. Lin, Z.-Y. Deng, J. Zheng, J. Cai, J.-C. Xiao, J. Fluorine Chem. 163 (2014) 38–41.
(e) F. Wang, L. Li, C. Ni, J. Hu, Beilstein J. Org. Chem. 10 (2014) 344–351.
(f) Y. Qiao, T. Si, M.-H. Yang, R.A. Altman, J. Org. Chem. 79 (2014) 7122–7131.

[7] S.R. Piettre, L. Cabanas, Tetrahedron Lett. 37 (1996) 5881–5884.

[8] (a) G.K.S. Prakash, Y. Wang, J. Hu, G.A. Olah, J. Fluorine Chem. 126 (2005) 1361–1367.

(b) Y. Zhao, W. Huang, L. Zhu, J. Hu, Org. Lett. 12 (2010) 1444–1447.

(c) X.-P. Wang, J.-H. Lin, J.-C. Xiao, X. Zheng, Eur. J. Org. Chem. (2014) 928–932.

[9] (a) T.M. Gøgsig, L.S. Søbjerg, A.T. Lindhardt, K.L. Jensen, T. Skrydstrup, J. Org. Chem. 73 (2008) 3404–3410.

(b) C. Ehm, F.A. Akkerman, D. Lentz, J. Fluorine Chem. 131 (2010) 1173–1181.

[10] (a) J. Ichikawa, M. Fujiwara, H. Nawata, T. Okauchi, T. Minami, Tetrahedron Lett. 37 (1996) 8799–8802.

(b) B.V. Nguyen, D.J. Burton, J. Org. Chem. 62 (1997) 7758–7764.

(c) J. Ichikawa, J. Fluorine Chem. 105 (2000) 257–263.

(d) A. Raghavanpillai, D.J. Burton, J. Org. Chem. 71 (2006) 194–201.

(e) J.H. Choi, I.H. Jeong, Tetrahedron Lett. 49 (2008) 952–955.

(f) F.A. Akkerman, R. Kickbusch, D. Lentz, Chem. Asian J. 3 (2008) 719–731.

(g) S.Y. Han, I.H. Jeong, Org. Lett. 12 (2010) 5518–5521.

(h) L. Lu, D.J. Burton, J. Fluorine Chem. 133 (2012) 16–19.

[11] (a) J.-P. Gillet, R. Sauvetre, J.-F. Normant, Tetrahedron Lett. 26 (1985) 3999–4002.

(b) J.P. Gillet, R. Sauvêtre, J.F. Normant, Synthesis (1986) 538–543.

[12] T. Fujita, T. Ichitsuka, K. Fuchibe, J. Ichikawa, Chem. Lett. 40 (2011) 986–988.

[13] For fluoroalkenylzinc reagents stabilized by TMEDA, see:

(a) B. Jiang, Y. Xu, J. Org. Chem. 56 (1991) 7336–7340.

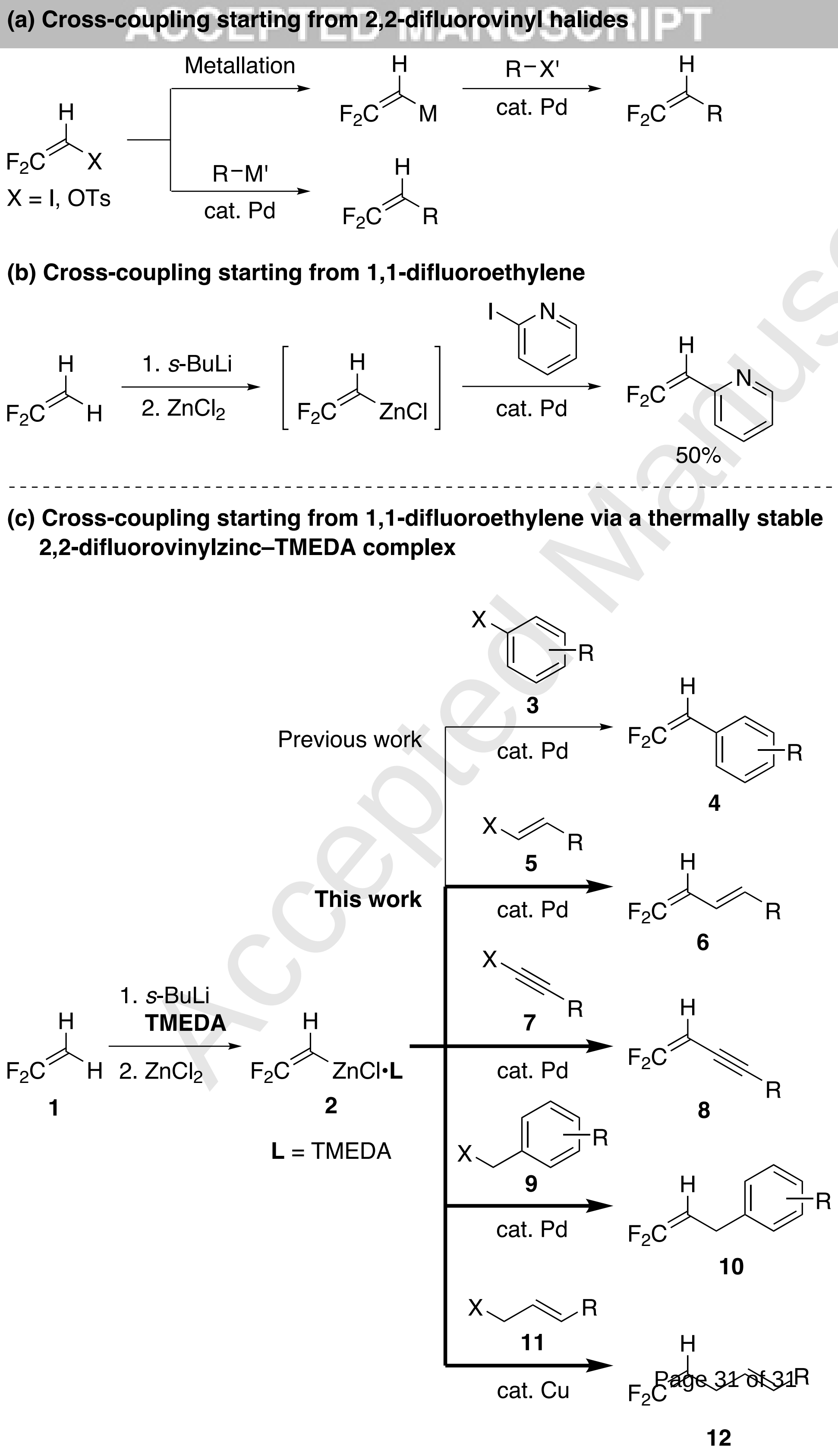
(b) A. Raghavanpillai, D.J. Burton, J. Org. Chem. 69 (2004) 7083–7091.

[14] For a review on the Negishi cross-coupling, see: E.-i. Negishi, Q. Hu, Z. Huang, M. Qian, G. Wang, Aldrichimica Acta 38 (2005) 71–88.

[15] M. Pohmakotr, K. Boonkitpattarakul, W. Ieawsuwan, S. Jarussophon, N. Duangdee, P. Tuchinda, V. Reutrakul, Tetrahedron 62 (2006) 5973–5985.

[16] I. Nowak, M.J. Robins, Org. Lett. 7 (2005) 721–724.

[17] M. Yokota, D. Fujita, J. Ichikawa, Org. Lett. 9 (2007) 4639–4642.


(a) Cross-coupling starting from 2,2-difluorovinyl halides
Metallation
R−X'
cat. Pd
R−M'
cat. Pd
X = I, OTs
(b) Cross-coupling starting from 1,1-difluoroethylene
1. s-BuLi
2. ZnCl2
cat. Pd
50%
(c) Cross-coupling starting from 1,1-difluoroethylene via a thermally stable 2,2-difluorovinylzinc–TMEDA complex
1. s-BuLi
TMEDA
2. ZnCl2
1
2
L = TMEDA
Previous work
This work
3
4
5
6
7
8
9
10
11
12
cat. Pd
cat. Cu